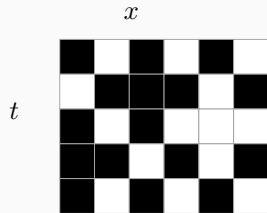


Relative-periodic domain selection by Bernoulli NML

1 Observed spacetime

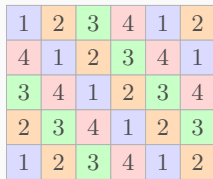
Choose candidate (p, s) .
Example: $p = 2, s = 1$.



Raw CA spacetime to explain. Each orbit contributes one background bit. Residual mask: $M = U \oplus B^*$. Output: periodic background + residual.

2 Orbit classes

Apply $(t, x) \mapsto (t + p, x + s)$.
Same colour = same orbit.



3 Majority-vote fit

Count ones and zeros in each orbit.
Choose the majority bit.

orbit	ones / zeros
1	5 / 1
2	1 / 5
3	4 / 2

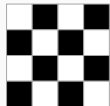
Output

one bit
per
orbit

4 Score and select

NML = fit + penalty.
Keep the best period.

$B^*(p, s)$


$$\sum_j n_j H_b(\hat{\theta}_j)$$

Fit

$$\frac{1}{2} \sum_j \log_2 n_j$$

Penalty

Select

Lowest NML wins.